Unit: Unit 0: Foundations of Algebra
Topic: GCF LCM

Name: ________________
Date: __________

Homework (Carolina Reaper)

Q1. Find the GCF of 12 and 20 using prime factorization.
(A) 2
(B) 4
(C) 6
(D) 8
(E) 12

Q2. A song has a rhythm pattern that repeats every 15 seconds. If another song has a pattern that repeats every 25 seconds, what is the LCM of their pattern lengths?
(A) 75
(B) 100
(C) 125
(D) 150
(E) 175

Q3. Find the LCM of 9 and 12 using multiples.
(A) 18
(B) 24
(C) 27
(D) 36
(E) 48

Q4. A painter uses a 12-inch brush and a 16-inch brush to paint a canvas. What is the GCF of the brush lengths in inches?
(A) 1
(B) 2
(C) 4
(D) 6
(E) 8

Q5. A music producer wants to mix two songs with frequencies of 12 Hz and 15 Hz. What is the LCM of their frequencies?
(A) 24
(B) 30
(C) 36
(D) 60
(E) 72

Q6. A canvas has a width of 18 inches and a length of 24 inches. What is the GCF of the dimensions in inches?
(A) 1
(B) 2
(C) 3
(D) 6
(E) 12

Q7. Find the GCF of 20 and 30 using factors.
(A) 5
(B) 10
(C) 15
(D) 20
(E) 25

Q8. A drummer has two rhythms with frequencies of 8 Hz and 12 Hz. What is the LCM of their frequencies?
(A) 16
(B) 20
(C) 24
(D) 32
(E) 40

Open Ended Questions (FRQ)

Q9. A painter is creating a mosaic with two different tile sizes: 12 inches and 18 inches. Find the GCF and LCM of the tile sizes and explain how they can be used to create a pattern that repeats.

Q10. A music producer is mixing a song with three different frequencies: 10 Hz, 12 Hz, and 15 Hz. Find the LCM of the frequencies and explain how it can be used to create a harmonious rhythm.

Chef's Tips

- Chef's Tips
- Ensure students understand the concept of GCF and LCM before applying it to real-world problems
- Encourage students to use prime factorization and multiples to find GCF and LCM